USER MANUAL & TUTORIAL

Cryobalance v0.1.0

Fast, Regularized Spherical Harmonic Orientation Balancing for Cryo-EM Metadata

CORE CAPABILITY

Cryobalance diagnoses preferred orientation bias in single-particle cryo-EM datasets and computes continuous, Tikhonov-regularized inverse-density weights over the unit sphere (S^2). Without deleting a single particle image, it restores 3D Fourier sampling sphericity directly through RELION STAR metadata.

1. Diagnostic Orientation Criteria

Before running lengthy 3D auto-refinements, use Cryobalance's rotationally invariant metrics to evaluate whether your grid suffers from severe air–water interface orientation clustering:

< 0.20	0.20 – 0.65	> 0.65
ISOTROPIC (IDEAL)	MODERATE BIAS	SEVERE MISSING CONE
OPTIMAL	BALANCING USEFUL	BALANCING CRITICAL

2. Fast Command-Line Workflow

The package provides a streamlined global CLI command that processes STAR files in seconds:

1. **Basic Execution (C1 Symmetry):**
   Compute harmonic density up to cutoff L_{max} = 12 and output weighted particles:

```
$ cryobalance run_data.star balanced_data.star
```

2. **Applying Symmetrized Expansion (e.g., C3 or D2):**
   Expands projections across all point-group rotation operators before density fitting:

```
$ cryobalance particles.star balanced.star --symmetry C3 --lmax 12 --alpha 0.02
```

3. **Direct 3D Reconstruction in RELION:**
   Back-project using native particle weights (`rlnParticleWeight`) without altering angles:

```
$ relion_reconstruct --i balanced.star --o balanced_map.mrc --ctf --sym c3
```

3. Python API Integration

Embed Cryobalance directly into custom preprocessing pipelines, automation daemons, or Jupyter Notebooks:

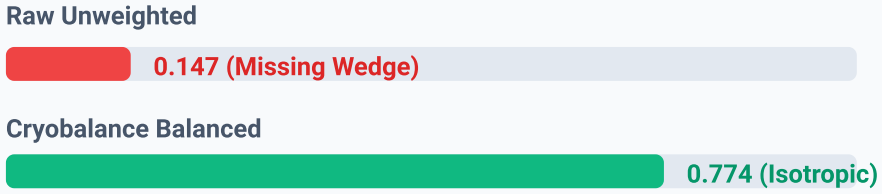
```
      import numpy as np import starfile from cryobalance.core import
compute_metrics_and_weights # 1. Load particle coordinates from RELION STAR file data =
starfile.read("particles.star") df = data["particles"] if isinstance(data, dict) else data
phi = np.deg2rad(df["rlnAngleRot"].to_numpy()) % (2.0 * np.pi) theta =
np.clip(np.deg2rad(df["rlnAngleTilt"].to_numpy()), 0.0, np.pi) # 2. Derive spherical
harmonics metrics and continuous weights weights, metrics = compute_metrics_and_weights(
phi=phi, theta=theta, l_max=12, sym="C3", alpha=0.02 ) # 3. Export balanced dataset for
downstream reconstruction df["rlnParticleWeight"] = np.round(weights, 5) starfile.write(data,
"balanced_particles.star", overwrite=True)
```

4. Quantitative Benchmark Performance

Benchmark on the canonical preferred orientation model (Influenza Hemagglutinin trimer, N = 60,000 particles):

Workflow	Symmetry	Anisotropy (A)	Effective Sample Size	3D Sphericity	Compute Time
Raw Unweighted	C1	24.3348	60,000 (100%)	0.147	Baseline
Cryobalance (C1)	C1	24.3348	14,152 (23.6%)	0.774	9.94 s
Cryobalance (C3)	C3	24.3335	14,149 (23.6%)	0.772	19.93 s

5. Reciprocal-Space Sphericity Recovery



AUTHOR & ACADEMIC CONTACT

Krishna Kant Gupta • Department of Bioinformatics, Rajiv Gandhi Institute of Information Technology and Biotechnology, Bharati Vidyapeeth (Deemed to be University), Pune 411046, India.
Email: krishnakant.gupta@gmail.com • GitHub: github.com/krishnakantgupta-ai/Cryobalance